

Andrey Zakharov

Full-Stack Software Engineer

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Full-Stack Software Engineer with 3+ years of hands-on project-based development experience. Builds end-to-end products across Python and Java backends, React interfaces, data/AI integrations, geospatial pipelines and simulation systems. Experienced in taking projects from architecture and MVP implementation to testing, packaging and deployment.

Experience

Independent Project Development: 2023 – Present.

COMPASS AI – Full-Stack & Applied AI Engineer

2026 – Present

Tech: Python, FastAPI, JavaScript, Streamlit, Pandas, NumPy, scikit-learn, PyTorch, ONNX, REST APIs, Plane API

- Architected and built a local full-stack decision-support system combining FastAPI services, browser interfaces, Streamlit dashboards, data pipelines and Plane project-management integration.
- Implemented CSV, JSON and Parquet import/export, interactive data views, Kanban workflows, bulk task assignment, reports and model-training workflows.
- Integrated six matching approaches and built ranked employee recommendations based on skills, workload, delivery risk, task complexity and development goals; evaluated ranking on 375 held-out tasks with ROC-AUC 0.859, MRR 0.944 and P@1 0.893.
- Added local Qwen/Ollama explanations, fairness reporting and modular orchestration for task analysis, team matching and recommendation generation.

UrbanFlow AI – Full-Stack Simulation Engineer

2026

Tech: Python, FastAPI, React, TypeScript, Three.js, Vite, SUMO, TraCI, OpenStreetMap, Pandas

- Developed a web platform that imports and normalizes OpenStreetMap data, generates SUMO scenarios and visualizes live microscopic traffic simulation in a React, TypeScript and Three.js interface.
- Integrated FastAPI, SUMO and TraCI to manage simulation sessions, traffic lights, road events, vehicles, pedestrians and public transport.
- Implemented interactive scenario editing, live metrics, congestion visualization, model registry and training-artifact tracking for experiments with 300–500 vehicles and hundreds to thousands of controlled traffic lights.

Face-Swap-Studio – Full-Stack AI Application Engineer

2025

Tech: Python, FastAPI, Pydantic, OpenCV, NumPy, InsightFace, ONNX Runtime, HTML/CSS/JavaScript, macOS

- Built a local full-stack AI application with a FastAPI backend and browser workflow for source/target image management, face detection, visual mapping, generation and result archives.
- Unified nine face-swap and head-model backends behind a common model registry with readiness checks, separate runners and support for multi-face and per-face assignments.
- Implemented single-image and batch processing with temporary-file isolation, safe downloads, optional GFPGAN restoration, Real-ESRGAN upscaling and macOS packaging.

Cartopia – Full-Stack & Geospatial Software Engineer

2023 – 2025

Tech: Java 17, Forge, Gradle, Jackson, Java HttpServer, HTML/CSS/JavaScript, GeoTIFF/DEM, OpenStreetMap, Open-Meteo

- Built a Java/Forge geospatial generation system that reconstructs real-world areas at approximately 1:1 scale, combining terrain, buildings, roads, railways, infrastructure, vegetation, time and weather data.
- Developed a local web interface and HTTP control server for place search, area selection, generation configuration, progress control and pipeline execution.
- Designed an extensible procedural-generation architecture with 80+ generator and storage classes, plus streaming terrain/geodata processing, validation, retries and cleanup for external data sources.

Skills

Technical: Python, FastAPI, REST APIs, React, TypeScript/JavaScript, HTML/CSS, Java, Gradle, Three.js, Vite, Streamlit, Pandas, NumPy, OpenCV, ONNX Runtime, SQL/SQLite, Git, Docker, OpenStreetMap, SUMO/TraCI.

Languages: English – C1 (Advanced); Russian – Native.

Education

SAINT PETERSBURG STATE UNIVERSITY OF AEROSPACE INSTRUMENTATION (SUAI) – Saint Petersburg, Russia

- Master's Degree, Management – Strategic Human Resource Management 2027
- Bachelor's Degree, Software Engineering – Software Systems Design 2025
- Professional Retraining Diploma – Web and Multimedia Applications Developer 2024
- Additional Professional Education: 250+ certificates and diplomas from 60+ universities, companies, educational platforms, institutes, and professional training programs worldwide, covering diverse academic and professional fields and disciplines.